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| 1 | <b>Course Name:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|   | Software Testing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 2 | <b>Course Code:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|   | ISB38104                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 3 | <b>Year and Semester Offered:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|   | Year 3 Semester 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4 | <b>Credit Hours Value:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|   | 4 Credit Hours [4 SLT Credit]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5 | <b>Prerequisite(s), if applicable:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|   | NIL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 6 | <b>Course Synopsis:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|   | This course introduces students to the fundamentals and methods of software testing. The course covers the different types of software testing, testing techniques, testing tools, testing methodologies, test case design, test execution and software testing best practices. The student will also gain hands-on experience with software testing tools and techniques through a series of practical exercises and a project.                                                                                              |
| 7 | <b>Course Learning Outcome:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|   | <p>Upon completion of the course, students should be able to:</p> <ol style="list-style-type: none"> <li>1. Explain testing concepts, the fundamentals of test processes, test approaches, and principles to support test objectives.</li> <li>2. Perform testing activities based on agreed test plans, analyze and report on the results of tests.</li> <li>3. Organize tests by using established testing techniques; analyze both functional and non-functional specifications at all test levels for systems.</li> </ol> |
| 8 | <b>Content Outline:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|   | <b>Topic</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|   | 1. Introduction To Software Testing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|   | 2. Fundamentals of Testing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|   | 3. Testing Throughout the Software Lifecycle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|   | 4. Static Testing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|   | 5. Dynamic Testing – Black Box                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|   | 6. Dynamic Testing – White Box                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|   | 7. Test Management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|   | 8. Tools                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|   | 9. Student's activities/Seminar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |